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Spatiotemporal Assessment and Correction of Gridded Precipitation Products in North Western Morocco

Latifa Ait Dhmane [1,*], Jalal Moustadraf [1], Mariame Rachdane [1], Mohamed Elmehdi Saidi [1], Khalid Benjmel [2], Fouad Amraoui [2], Mohamed Abdellah Ezzaouini [3], Abdelaziz Ait Sliman [4] and Abdessamad Hadri [5]

[1] Geo-Resources, Geo-Environment and Civil Engineering Laboratory, Cadi Ayyad University, Marrakesh 40000, Morocco; j.moustadraf@uca.ac.ma (J.M.); mariame.rachdane@ced.uca.ma (M.R.); m.saidi@uca.ma (M.E.S.)
[2] Laboratory of Geosciences Applied to Engineering Development (GAIA), Faculty of Science Ain-Chock, Hassan II University, Casablanca 20150, Morocco; khalid.benjmel-etu@etu.univh2c.ma (K.B.); fouad.amraoui@univh2c.ma (F.A.)
[3] Loukkos Basin Hydraulic Agency, Tetouan 93000, Morocco; ezzaouini70@yahoo.fr
[4] Hydroleader Office, Marrakech 40000, Morocco; aziz.sliman@gmail.com
[5] International Water Research Institute, Mohammed VI Polytechnic University (UM6P), Benguerir 43150, Morocco; abdessamad.hadri@um6p.ma
* Correspondence: latifa.aitdhmane-etu@etu.univh2c.ma

Abstract: Accurate and spatially distributed precipitation data are fundamental to effective water resource management. In Morocco, as in other arid and semi-arid regions, precipitation exhibits significant spatial and temporal variability. Indeed, there is an intra- and inter-annual variability and the northwest is rainier than the rest of the country. In the Bouregreg watershed, this irregularity, along with a sparse gauge network, poses a major challenge for water resource management. In this context, remote sensing data could provide a viable alternative. This study aims precisely to evaluate the performance of four gridded daily precipitation products: three IMERG-V06 datasets (GPM-F, GPM-L, and GPM-E) and a reanalysis product (ERA5). The evaluation is conducted using 11 rain gauge stations over a 20-year period (2000–2020) on various temporal scales (daily, monthly, seasonal, and annual) using a pixel-to-point approach, employing different classification and regression metrics of machine learning. According to the findings, the GPM products showed high accuracy with a low margin of error in terms of bias, RMSE, and MAE. However, it was observed that ERA5 outperformed the GPM products in identifying spatial precipitation patterns and demonstrated a stronger correlation. The evaluation results also showed that the gridded precipitation products performed better during the summer months for seasonal assessment, with relatively lower accuracy and higher biases during rainy months. Furthermore, these gridded products showed excellent performance in capturing different precipitation intensities, with the highest accuracy observed for light rain. This is particularly important for arid and semi-arid regions where most precipitation falls under the low-intensity category. Although gridded precipitation estimates provide global coverage at high spatiotemporal resolutions, their accuracy is currently insufficient and would require improvement. To address this, we employed an artificial neural network (ANN) model for bias correction and enhancing raw precipitation estimates from the GPM-F product. The results indicated a slight increase in the correlation coefficient and a significant reduction in biases, RMSE, and MAE. Consequently, this research currently supports the applicability of GPM-F data in North Western Morocco.

Keywords: Morocco; gridded precipitation products; GPM; ERA5; sparse gauge network; artificial neural network

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1. Introduction

The issue of water availability and access is undoubtedly one of the major problems that humanity will face in the coming century. This problem is particularly alarming in the

Mediterranean region, which is recognized as one of the most sensitive areas to climate change [1]. Climate models predict a decreasing trend in Mediterranean precipitation over the next few decades [2,3]. In this context, ensuring access to water poses a significant challenge for sustainable development, with a clear environmental dimension that involves preserving a threatened resource in terms of both quantity and quality [4,5]. Water plays a crucial role in socio-economic development, and accurately assessing its availability is a primary concern for governments, particularly in countries heavily reliant on agricultural revenue [5,6].

Conventional methods for measuring precipitation primarily rely on rain gauges and weather radars. However, these traditional approaches have limitations in terms of data quality, maintenance costs, and potential data quality issues [7,8]. With advancements in numerical simulations and Earth observation techniques, satellite-based and model-based methods have emerged as effective approaches for collecting accurate precipitation data with consistent spatiotemporal coverage. Satellite-based precipitation estimation products utilize satellite-enhanced meteorological observations, enabling estimation through the joint inversion of visible, infrared, microwave, or multi-sensor data [9,10]. On the other hand, the reanalysis-based approach combines ground-based observation data with climate model forecasts using a data assimilation system to estimate surface precipitation [11,12].

The choice of the Integrated Multi-Satellite Retrievals for GPM (IMERG) is justified by its ability to integrate multi-satellite sensors including passive microwave, infrared, and radar, its high spatiotemporal resolution, and its near real-time access. Also, it incorporates advanced data assimilation techniques that integrate satellite observations with other available data sources, such as gauge observations and numerical weather prediction models [13]. This assimilation process improves the accuracy and reliability of precipitation estimates. Furthermore, the strengths of the Fifth Generation of ECMWF Atmospheric Reanalysis (ERA5) lie in its long-term dataset and the use of rigorous quality control procedures to filter out erroneous or inconsistent observations. ERA5 also utilizes state-of-the-art numerical weather prediction models developed by ECMWF. These models integrate various atmospheric variables, including temperature, humidity, wind, and pressure, to generate consistent and physically based precipitation estimates [14]. These strengths make GPM and ERA5 a valuable resource for estimating and studying precipitation.

Data from IMERG and ERA5 have been extensively examined and compared to observed data, both globally [7,15–18] and at the national scale as in sub-Saharan Morocco [19] or in the Oum Er Rbia watershed [20,21]. These data have also been used for flood simulation [22,23] and for climate change studies and drought monitoring in the Tensift sub-basin [24,25]. However, studies evaluating the effectiveness of precipitation products in the Bouregreg watershed, in northwestern Morocco, are currently limited. The objective of this study is then to assess the reliability and accuracy of the IMERG V06 and ERA5 precipitation products for estimating precipitation in this region using machine learning metrics for classification and regression. Choosing the appropriate metric for evaluating a precipitation product's estimation capability is crucial. In a review conducted by the World Meteorological Organization [26], various evaluation scores were examined, including the correlation coefficient, the mean absolute error (MAE), the root mean square error (RMSE), the probability of detection (POD), and the false alarm rate (FAR). These parameters enable the measurement of both the precision and detection capability of the precipitation product [27]. This study considered these parameters, along with other machine learning metrics. Furthermore, although gridded precipitation products (GPPs) offer the advantages of global coverage and high spatiotemporal resolution [28], their use in hydrological applications is sometimes limited due to associated biases. These products are indeed susceptible to inherent errors, such as errors related to precipitation measurement [29], variations in land surface properties [30,31], and variations in climate regimes, seasons, and elevations [32–34]. To address these concerns, this study proposes, in addition to the evaluation of gridded precipitation data, a framework to correct the biases of these precipitations.

2. Materials and Methods

2.1. Study Area and Rain Gauge Datasets

This study focuses on the Bouregreg watershed, which is situated in the central-western part of Morocco, spanning latitudes 32.82° N to 34.04° N and longitudes 5.48° W to 6.86° W (Figure 1). The watershed is bordered by the Saiss and Ghareb ditches to the north, the phosphate plateau to the south, the Middle Atlas Mountains to the east, and the Atlantic Ocean to the west. Administratively, it encompasses provinces from the Rabat-Salé-Kenitra and Beni Mellal-Khenifra regions. The total area of the watershed is approximately 9462 km^2, consisting of four sub-basins. The hydrographic network is formed by four main rivers: Bouregreg (264 km), Grou (278 km), Medior (94 km), and Korifla (110 km). Geomorphologically, the area comprises three natural units: the Middle Atlas Mountains (covering the upper Bouregreg), the Oulmès-Zaêrs plateau (encompassing the middle Bouregreg), and the Chaouia plain [35]. The climate of the study area is semi-arid to sub-humid. The average annual rainfall ranges from 350 mm in the plain to nearly 600 mm in the mountainous zone. The monthly distribution of precipitation shows two distinct seasons: (a) a rainy season from October to April, accounting for the majority of the annual rainfall; (b) a dry season from May to September. Average temperatures in the area range from 12 °C to 18 °C in the mountainous zone and to around 20 °C in the Bouregreg plain.

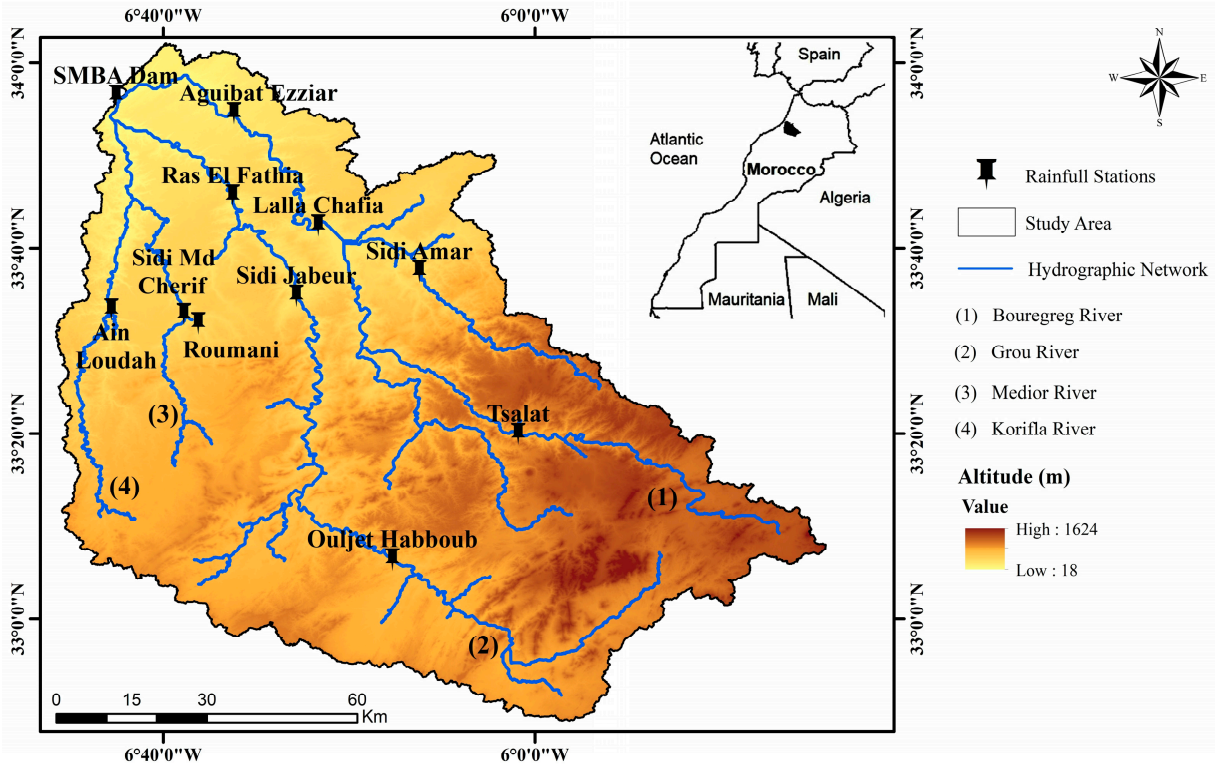


Figure 1. Geographical location of the Bouregreg watershed and rain gauge network.

In this study, daily data from 11 rain gauge stations, provided by the Bouregreg and Chaouia Hydraulic Basin Agency (ABHBC), were used to assess the GPPs. To ensure the reliability of the evaluation, a threshold of 20% missing data was set, beyond which the records were not considered. The threshold for excluding missing values was justified by referring to recent research works [36–38]. These authors defined a threshold of 20% of missing data not to be exceeded to ensure the reliability of the assessment. Therefore, to avoid the consequences and impact of missing values on the Sidi Amar rain gauge and other rain gauges, as well as to preserve the integrity of the assessment and ensure its reliability, the missing days in each station were excluded from both ground stations and satellite products.

Table 1 provides the coordinates of the used stations on a daily time step, along with the percentage of missing values.

Table 1. Station coordinates used in this study and their rate of missing values.

Station Name	Longitude W	Latitude N	Z(m)	Missing Values (%)
Aguibat Ezziar	−6.54	33.91	81	0.11
Ain Loudah	−6.76	33.56	163	0.08
SMBA Dam	−6.75	33.94	67	1.31
Lalla Chafia	−6.39	33.71	193	0.08
Ouljet Habboub	−6.26	33.11	557	2.93
Ras El Fathia	−6.54	33.76	132	0.07
Rommani	−6.60	33.53	308	0.12
Sidi Amar	−6.21	33.63	263	16.61
Sidi Jabeur	−6.43	33.58	197	0.07
Sidi Md Cherif	−6.63	33.55	282	0.10
Tsalat	−6.03	33.35	687	0.12

The spatial distribution of the 11 rain gauges is characterized by their concentration downstream of the watershed. Indeed, nine stations are located below 300 m, and the two highest rain gauges are situated at 557 m (Ouljet Haboub) and 687 m (Tsalat), respectively. The higher regions, up to the peak at 1624 m, lack ground observation stations. In this context of sparse ground-based observations, satellite products will be particularly useful.

2.2. The Gridded Precipitation Products

In order to estimate and spatialize precipitation amounts, satellites are equipped with specialized sensors that capture and measure various properties of clouds and precipitation systems from space. These measurements are then processed using sophisticated algorithms to estimate precipitation rates and distributions over vast areas. However, it is important to validate and calibrate these estimates in relation to ground observations and potentially correct their biases [39].

ERA5 is the latest version of atmospheric reanalysis data provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). It covers the period from 1950 onwards and it replaced the previous version, ERA-Interim (spanning 1979 onwards), that was implemented in 2006 and covered the period from 1979 [14]. ERA5 was developed based on the integrated forecasting system Cy41r2, which has been in operation since 2016, benefiting from years of work in model physics, core dynamics, and data assimilation. The ERA5 dataset has a spatial resolution of 0.25° and a temporal resolution of 1 h. In contrast, ERA5-Land focuses specifically on land data and provides a more detailed characterization of land conditions with an enhanced resolution of 0.1°. The main difference between ERA5 and ERA5-Land is that the latter is influenced only by ERA5's lower atmospheric weather field, with an additional correction for evaporation rate to improve accessibility [40]. In this study, the advanced ERA5 dataset is used.

Our analysis also utilized three versions of the IMERG V06 products. IMERG is a collection of satellite-based precipitation data provided by NASA, offering a fine spatial resolution of 0.1° and a temporal resolution of 0.5 h [13]. It builds upon the previous TMPA services, covering a wider spatial range from 60° N to 60° S. The IMERG product consists of three runs: Early (ER), Late (LR), and Final (FR). The ER and LR are near-real-time products released 4 h and 12 h after observations, respectively. They differ in the morphing technique used: forward morphing for ER and both forward and backward morphing for LR. The FR, which is the research-level product, is calibrated using monthly GPCC (Global Precipitation Climatology Centre) data, providing more accurate precipitation information [7]. The precipitation estimates from these satellite data encompass various forms of precipitation, including rain, drizzle, snow, graupel, and hail [41].

2.3. Methodology

This section outlines the methodology employed to assess the performance of the GPPs at different temporal scales. Data spanning from September 2000 to August 2020 were analyzed by directly comparing them to point rain gauge measurements, utilizing machine learning techniques such as classification and regression. For the spatial scale, we obviously chose the pixels covering the rain gauges. In this regard, it is worth noting that interpolation methods do not guarantee an accurate representation of precipitation distribution due to the limited spatial density of rain gauges [42]. Moreover, a bias correction framework was introduced for the GPM-F precipitation product, employing the artificial neural network (ANN) approach. The framework encompasses three modules: Construction and Training, Testing, and Precipitation Correction (Figure 2).

2.3.1. Classification Metrics

In classification, the objective is to predict the outcome from a finite set of categorical values [43]. Simply put, a classification metric quantifies how well a machine learning model assigns observations to specific classes [44]. Classification metrics can be categorized into three main groups: binary, multiclass, and multilabel.

The confusion matrix, also known as an error matrix, is a specific tabular representation that provides a visual depiction of a model's performance [45,46]. As the name implies, it is an NxN matrix (Table 2) [47], where N represents the number of predicted classes. The confusion matrix consists of two dimensions, "Actual" and "Predicted", with "Classes" represented along each dimension. Various performance measures are computed based on the values in the confusion matrix. When evaluating the performance of a precipitation estimation model from a classification perspective, there are four possible outcomes:

(a) True Positive (TP): this indicates the number of days when both the ground observation and the satellite predict the occurrence of rainfall.
(b) True Negative (TN): this parameter signifies the absence of rainfall, as neither the satellite nor the observation recorded any precipitation.
(c) False Positive (Type I Error) (FP): this denotes instances where the satellite incorrectly identifies a rainfall event that is not present in the observed data.
(d) False Negative (Type II Error) (FN): this represents the number of days when the satellite fails to detect a rainfall event that is recorded by the rain gauge.

Table 2. Confusion matrix for binary classification.

		Actual Value	
		Positive	**Negative**
Predicted Value	Positive	TP (True Positive)	FP (False Positive)
	Negative	FN (False Negative)	TN (True Negative)

Classification accuracy (ACC) is a widely used metric for evaluating a model's performance. It measures the ratio of correct predictions to the total number of predictions made on a dataset. ACC provides an overall success rate, where a perfect score of 1 indicates zero false positives (FP) and false negatives (FN) [48]. Conversely, classification error (Err) represents the ratio of incorrect predictions to the total number of predictions, reflecting the proportion of incorrect estimates made by the model [49].

Each prediction made by a product to determine the true class of the target contributes to an error. This results in false positives (FP) and false negatives (FN), indicating misclassifications compared to the true class. The precision, also known as the positive predictive value (PPV), focuses on minimizing false positives or false alarms. The precision is defined as the fraction of all positive predictions that are true positives [48,50]. Recall, also known as the true positive rate (TPR) or sensitivity, represents the number of true positives divided by the sum of true positives and false negatives. In other words, recall captures the fraction

of all actual positives that are correctly predicted as positives [50,51]. High recall with low precision indicates that most positive examples are correctly identified (low number of false negatives), but there are also many false positives. Conversely, low recall with high precision suggests that many positive examples are missed (high false negatives), but the ones predicted as positive are indeed positive (low number of false positives).

There exists a trade-off between precision and recall, making it challenging to optimize both simultaneously. Improving precision, which focuses on reducing false positives, often leads to a deterioration in recall, which concerns false negatives, and vice versa. This raises the question of which metric should be prioritized: precision or recall? To strike a balance between the two, when evaluating the effectiveness of a classification model in predicting positive instances, the F1 score is commonly used. The F1 score is calculated as the harmonic mean of precision and recall [51,52] and ranges between 0 and 1. It represents a compromise between precision and recall for assessing a model's performance [53]. The F1 score formula can be simplified and directly expressed using the components of the confusion matrix. It compares the true positive predictions (TP) with the model errors (FN + FP). An F1 score of 50% indicates that precision and recall are equal, which can occur when the model produces an equal number of false positives and false negatives.

The AUC-ROC (Area Under the Curve of the Receiver Operating Characteristic) curve is a valuable tool for visually assessing the performance of a classification model [54]. The ROC curve is constructed by plotting the true positive rate (TPR) against the false positive rate (FPR), with TPR on the *y*-axis and FPR on the *x*-axis. The AUC value represents the area under this curve, and a higher AUC value closer to 1 indicates superior model performance. The AUC-ROC curve is commonly used in binary classification problems [55]. TPR, also known as sensitivity, measures the proportion of positive examples correctly identified by the model, while specificity measures the proportion of negative examples correctly identified. By varying the decision threshold, the ROC curve evaluates the model's ability to distinguish between positive and negative classes [56].

The false discovery rate (FDR) quantifies the proportion of false positive predictions among all positive predictions made by the model. It provides a measure of the fraction of incorrect estimates within all the model's predictions [57] (Table 3).

Table 3. Summary of classification metric formulas.

$\text{ACC} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}}$	$\text{Specificity} = \frac{\text{TN}}{\text{TN}+\text{FP}}$
$\text{Err} = \frac{\text{FP}+\text{FN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} = 1 - \text{ACC}$	$\text{FPR} = 1 - \text{Specificity} = \frac{\text{FP}}{\text{FP}+\text{TN}}$
$\text{Precision} = \frac{\text{TP}}{\text{TP}+\text{FP}}$	$\text{FDR} = \frac{\text{FP}}{\text{FP}+\text{TP}}$
$\text{Recall} = \frac{\text{TP}}{\text{TP}+\text{FN}}$	$\text{F1_Score} = \frac{2\ (\text{Precision} \times \text{Recall})}{(\text{Precision}+\text{Recall})} = \frac{2\ \text{TP}}{2\ \text{TP}+\text{FP}+\text{FN}}$

2.3.2. Regression Metrics

In regression, the objective is to predict a continuous value rather than a class label [43]. Unlike classification, where accuracy is measured by predicting the correct class label, regression focuses on the closeness of the prediction to the actual value [58]. Commonly used regression metrics in machine learning include R^2 score (coefficient of determination) and Pearson correlation coefficient, as well as root mean squared error (RMSE) and mean absolute error (MAE).

The R^2 score evaluates the model's prediction quality in linear regression by assessing its performance relative to the variance in the data. It provides an indication of how well the satellite data correspond to the observed data. RMSE and MAE serve similar purposes, measuring the average magnitude of the error between estimated and observed precipitation values [59]. RMSE gives more weight to larger errors compared to MAE. Squaring the errors in RMSE amplifies the effect of larger errors, which can be useful in situations where large errors are particularly undesirable. This means that the RMSE is sensitive to outliers, while MAE is less affected by extreme values [60]. RMSE is often

considered a superior metric to MAE when the majority of prediction errors are small and outliers are infrequent or insignificant. However, when the distribution of errors is highly skewed or when the impact of outliers needs to be minimized, MAE might be a more suitable choice. The same when the magnitude of the error is more important than its squared value. In this case, MAE can provide a more intuitive understanding of the model's performance.

Bias is another commonly used metric for evaluating model performance. Negative bias values indicate underestimation of precipitation, while positive values indicate overestimation [19]. The Pearson correlation coefficient [61] assesses the goodness of fit and linear association between satellite-based precipitation products and observed precipitation. It ranges from 0 to 1, with 1 indicating a perfect fit.

For spatial analysis and a better understanding of the product's ability to capture spatial precipitation patterns, the Spearman spatial correlation coefficient was used to assess the consistency of spatial distributions between the products and stations [17]. This coefficient, based on rank correlation, provides a measure of similarity in spatial trends. It is calculated at different temporal scales (daily, monthly, and yearly) to evaluate the products' performance (Table 4).

Table 4. Summary of regression metric formulas.

$R^2 = 1 - \frac{\sum_{i=1}^{N}(X_i - Y_i)^2}{\sum_{i=1}^{N}(X_i - \overline{X})^2}$	$Bias = \frac{1}{N}\sum_{i=1}^{N} Y_i - X_i$
$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(X_i - Y_i)^2}$	$Pearson\ Corr = \frac{\sum_{i=1}^{N}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum_{i=1}^{N}(X_i - \overline{X})^2} \times \sqrt{\sum_{i=1}^{N}(Y_i - \overline{Y})^2}}$
$MAE = \frac{1}{N}\sum_{i=1}^{N} \mid X_i - Y_i \mid$	$Spearman\ Corr = 1 - \frac{(r_i - s_i)^2}{N^3 - N}$

Where X_i: gauge observation; Y_i: satellite precipitation estimation; $\overline{X}$: mean gauge observation; $\overline{Y}$: satellite mean precipitation estimation; r_i and s_i indicate the sequence numbers of the product and reference records, respectively, after sorting them in ascending order based on precipitation amount; and N: the sample size.

2.3.3. Precipitation Intensity Classes

In addition to assessing overall performance, an evaluation has been carried out specifically for different precipitation intensities. To categorize precipitation intensity, we utilized five classes based on the World Meteorological Organization guidelines [62]. These classes were further modified and tailored to suit the specific climate conditions of our study area (Table 5).

Table 5. Ranking of rainfall events based on their daily intensity. [a, b[: The interval includes values greater than or equal to a and strictly less than b.

Rainfall Event	Intensity in mm/d
light rain	[0, 1[
moderate rain	[1, 5[
medium rain	[5, 20[
heavy rain	[20, 40[
extreme rain	≥40

2.3.4. The Pixel Scale

In order to compare precipitation products with different spatial resolutions, previous studies have employed interpolation techniques to match ground observations onto grids for pixel-by-pixel validation [63,64]. However, due to limited station coverage and their dispersed locations within the Bouregreg basin, using station interpolation data alone may not accurately represent the actual precipitation conditions [65,66]. Therefore, precipitation estimates of the pixel corresponding to the gauge station location were extracted, enabling a pixel-to-point evaluation [67,68]. Moreover, previous studies have compared pixel-to-

pixel and pixel-to-point interpolation techniques and have found that they lead to similar conclusions [69,70].

2.3.5. Correction of Precipitation Estimates

The spatial variability of precipitation in the Bouregreg watershed has been the subject of few analyses [71]. However, the sparse and uneven distribution of rain gauges raises concerns about the reliability of spatial interpolation methods such as inverse distance weighting and ordinary kriging [72]. Bias correction is a vital step in improving the accuracy and reliability of satellite precipitation estimates. The latter is indeed prone to systematic errors or biases that can impact the quality of the estimates. Bias correction techniques then aim to minimize these biases and bring the satellite-derived estimates in closer agreement with ground-based observations. In this regard, an advanced bias correction method will be applied to the most suitable model using artificial neural networks (ANN) for regression. ANN is a parallel distributed information processing system that exhibits characteristics similar to the biological neural networks in the human brain [73,74]. The concept of artificial neurons was initially proposed by McCulloch and Pitts in 1943 [75], and ANN does not rely on assumptions about variables, making it suitable for complex and unstructured problems. It is also robust to specification errors and serves as a parsimonious universal approximator [76–78].

The constructed model is a linear stack of layers. Specifically, a sequential model of Keras. The first layer is a fully connected dense layer that has 64 neurons with a readback activation function and a predefined input shape. The next layers have 128, 256, and 128 neurons, respectively, with a relu activation function and which are hidden layers that nonlinearly transform input into output. The final layer is also a dense layer, but with a single neuron and no activation function specified, making it an output layer for a regression. The model uses a backpropagation learning approach to minimize a loss function that quantifies the difference between the predicted values and the actual values of the dependent variable. More precisely, for observation i, the prediction $\hat{y}_i$ is obtained using the formula below. The mean squared error (MSE) is used as the loss function, which calculates the mean of the squares of the differences between the predictions and the actual values. The loss function is minimized by the Adam optimization algorithm, which adjusts weights and biases during training using error backpropagation. In summary, the model combines matrix multiplication, addition, nonlinear activation, and optimization techniques to learn a function that predicts the dependent variable given the independent variable.

The tuning or training parameters include the number of epochs, which represents the total number of iterations performed by the model on the entire training data. Each epoch allows traversing the entire data once. In our model, we used 100 epochs, meaning that the model will be trained for 100 complete iterations. The batch_size parameter corresponds to the number of training samples used in a single iteration during the optimization of the model's weights. The training data are divided into batches, and each batch is used to update the model's weights. In this model, we have set batch_size = 32, indicating that each iteration will use a batch of 32 samples.

$$\hat{y}_i = f(w^T x_i + b) \qquad MSE = \frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{y}_i)^2$$

where: x_i is the input vector; w is a weight vector; b is the bias (a constant) learned by the model during training; f is the activation function of each layer; N is the total number of observations in the training dataset; Y_i is the actual value of the dependent variable for observation i; and $\hat{y}_i$ is the model prediction for observation i.

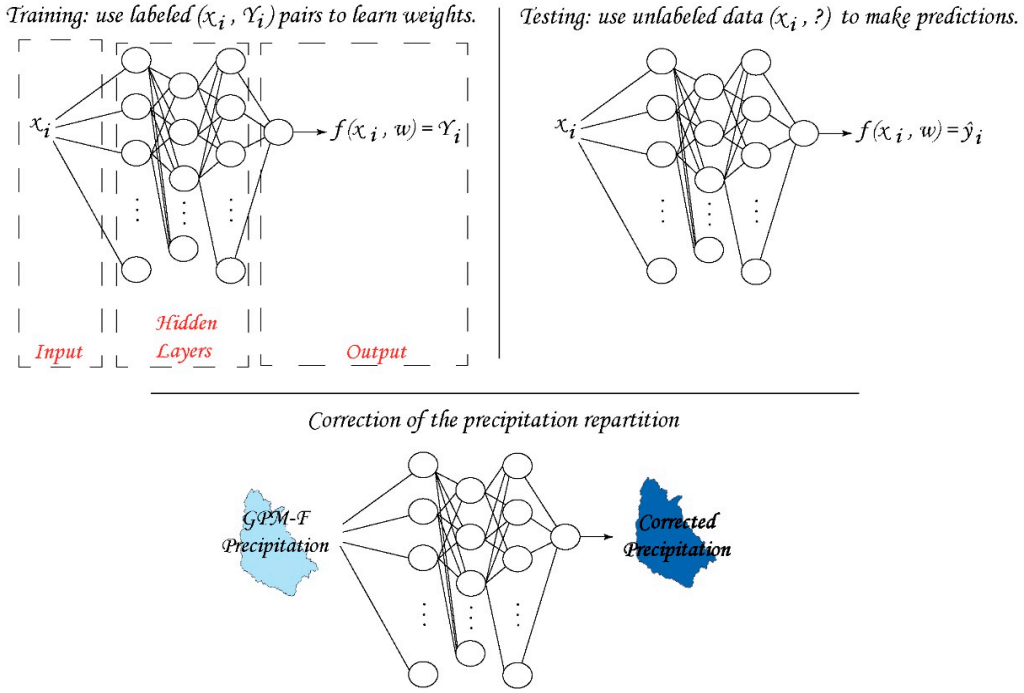


Figure 2. Training and testing in the neural network context ("?" means the predicted value).

3. Results and Discussion

3.1. Pixel Scale

The results show that the accuracy values for the GPM-L and GPM-F models are 0.75 and 0.74, respectively, indicating relatively good performance. However, the ERA5 model has a lower accuracy value of 0.46. For the precision, GPM-F and GPM-L have median values of 0.38, while GPM-E has a precision value of 0.37. The ERA5 model has a lower value of 0.22, indicating a relatively low ability to predict positive data. Overall, these precision values suggest that the GPM products have a slightly better performance than ERA5 in terms of predicting positive data, but all models may need improvements to be more accurate (Figure 3). Concerning recall values, the median for GPM-F and GPM-L is 0.81, while GPM-E has a value of 0.80. The ERA5 model has a higher recall value of 0.98, indicating a better ability to detect positive data, but at the cost of a higher number of false alarms. In general, high recall values indicate a good ability to detect positive data. However, it is important to consider recall in conjunction with precision to obtain a complete picture of the model's performance. In this regard, the F1-score combines precision and recall to evaluate the model's binary classification performance. The median F1-score for GPM-F and GPM-L is 0.52, while GPM-E has an F1-score of 0.50. The ERA5 model has a lower F1-score of 0.36. The AUC-ROC values indicate the model's ability to distinguish between positive and negative classes. The GPM-F and GPM-L products have AUC-ROC values of 0.78, while GPM-E has a value of 0.77. The ERA5 model has a lower AUC-ROC value of 0.67, suggesting the lowest performance in this aspect. For the false discovery rate (FDR), which evaluates the proportion of errors among positive predictions, GPM-F and GPM-L have an FDR of 0.62, while GPM-E has a value of 0.63. The ERA5 model has a higher FDR of 0.78, indicating a higher rate of false positive predictions.

Overall, and in conclusion, the GPM products generally perform better than the ERA5 model in terms of accuracy, precision, F1-score, AUC-ROC, and FDR (Figure 3). The good performance of the GPM products is probably due to its relatively high spatial resolution compared to the ERA5 reanalysis product.

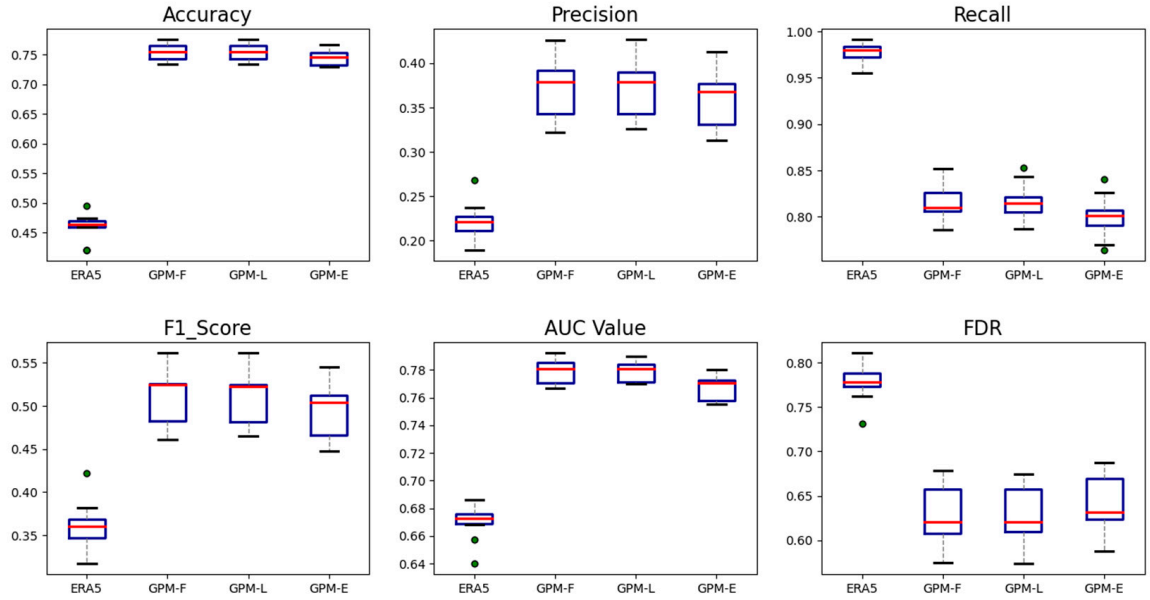


Figure 3. Boxplots of classification metrics at the daily scale of IMERG and ERA5 precipitation products (Red line: Median; Dots: Outliers).

The regression metrics indicate that at all temporal scales (daily, monthly, and annual), the GPM products (GPM-F, GPM-L, and GPM-E) generally exhibit lower bias compared to the ERA5. The latter tends to overestimate precipitation, while the GPM-F consistently demonstrates the lowest bias among the GPM products. Regarding correlation, the results show that the GPM products have improved correlations with observed data at the monthly and annual scales compared to the daily scale. At this daily scale, the ERA5 product exhibits the highest correlation, followed by GPM-F, GPM-L, and GPM-E. However, on the monthly scale, GPM-F shows the highest correlation, followed by ERA5, GPM-L, and GPM-E. The correlation improvement is more significant at the monthly scale than at the annual scale (Figure 4).

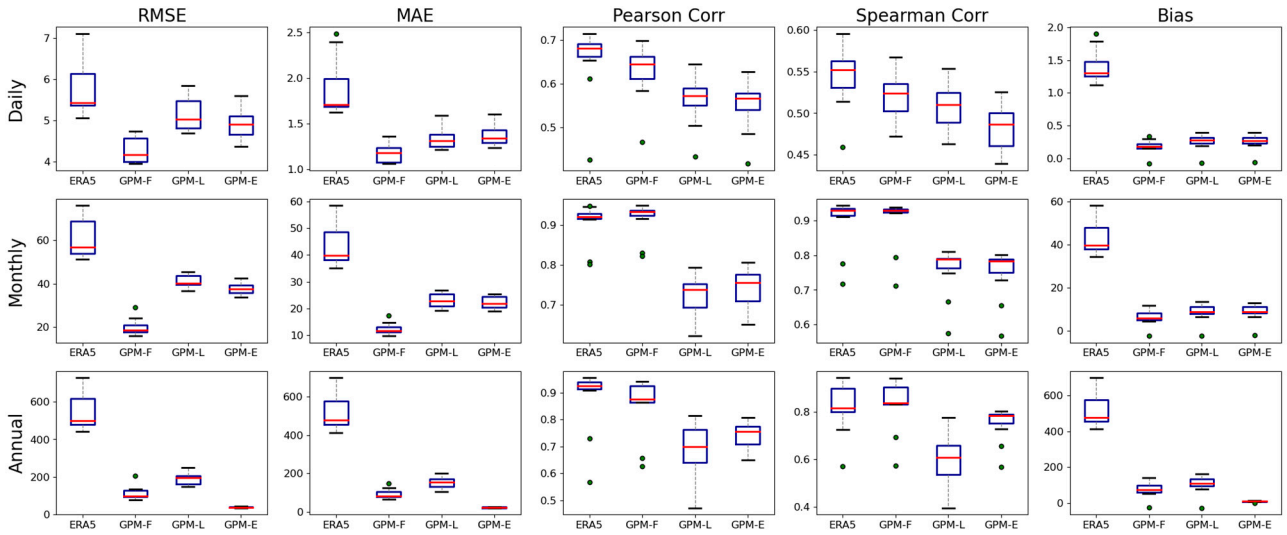


Figure 4. Boxplots of regression metrics on daily, monthly, and annual time scales of IMERG and ERA5 precipitation products (Red line: Median; Dots: Outliers).

In terms of spatial consistency, the ERA5 and GPM-F products exhibit the most consistent spatial patterns with the station data across all temporal scales. On the daily scale, ERA5 performs better than GPM-F, while on the monthly and annual scales, GPM-F

outperforms ERA5. GPM-L and GPM-E also show relatively good performance at the monthly and annual scales but weaker performance at the daily scale. Ref. [17] also obtained similar results in their evaluation of IMERG and ERA5 precipitation products' performance over the Mongolian Plateau. Additionally, the root mean square error (RMSE) and mean absolute error (MAE) values show that the GPM-F product performs well with the lowest RMSE and MAE values at all temporal scales. GPM-L and GPM-E products also outperform the ERA5 model in terms of error metrics. The ERA5 product consistently ranks last in terms of error and bias across all temporal scales (Figure 4).

These findings suggest that the GPM products, particularly GPM-F, demonstrate better performance in terms of bias, correlation, and error metrics compared to the ERA5 model. However, it is important to consider the specific context, dataset, and evaluation criteria when interpreting and generalizing these results.

The coefficient of determination (R^2) is a measure of how well the regression model fits the observed data. The confidence intervals, depicted by shaded regions, provide a range of potential values for both the slope and intercept of the regression line. These intervals help visualize the uncertainty associated with these parameters. The provided results indicate that at the daily scale, the GPM-F, GPM-L, and GPM-E models have relatively low R^2 values, suggesting that these models are not very effective in explaining the variance of the observed data at this time scale. In contrast, the ERA5 model has a higher R^2 value, indicating a better correspondence between the observed data and the linear regression model (Figure 5). However, at the monthly and annual scales, all four products show higher R^2 values, indicating that they are more effective in explaining the variance of the observed data. The GPM-L, GPM-E, and GPM-F models demonstrate R^2 values above 0.5 on the monthly scale, with the highest value of 0.87 for GPM-F. Similarly, at the annual scale, all R^2 values are relatively high, with values ranging from 0.49 to 0.86. These results suggest that the products have a better ability to explain the variance of the observed data at the monthly and annual scales compared to the daily scale (Figures 6 and 7). Moreover, the results indicate that the improvement in R^2 is more significant at the monthly scale than at other temporal scales, the ERA5 and GPM-F models exhibit good fits to the observed data. The performance of GPM-L and GPM-E is relatively acceptable at the monthly and annual scales but weaker at the daily scale.

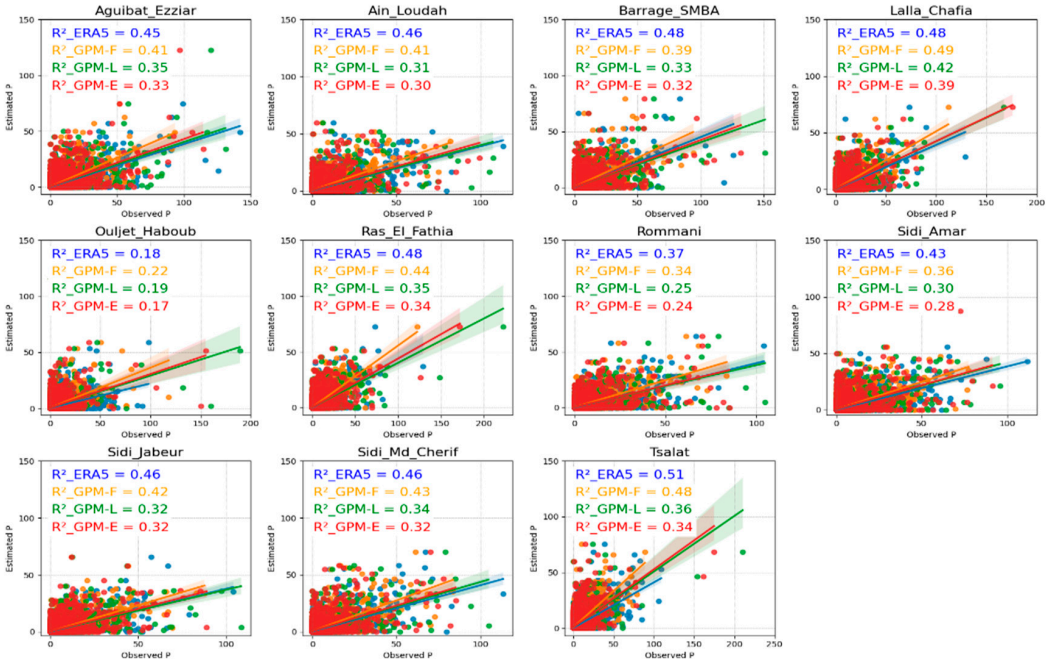


Figure 5. Linear regression between the daily precipitation observed by the rain gauges and that estimated by the IMERG and ERA5 precipitation products.

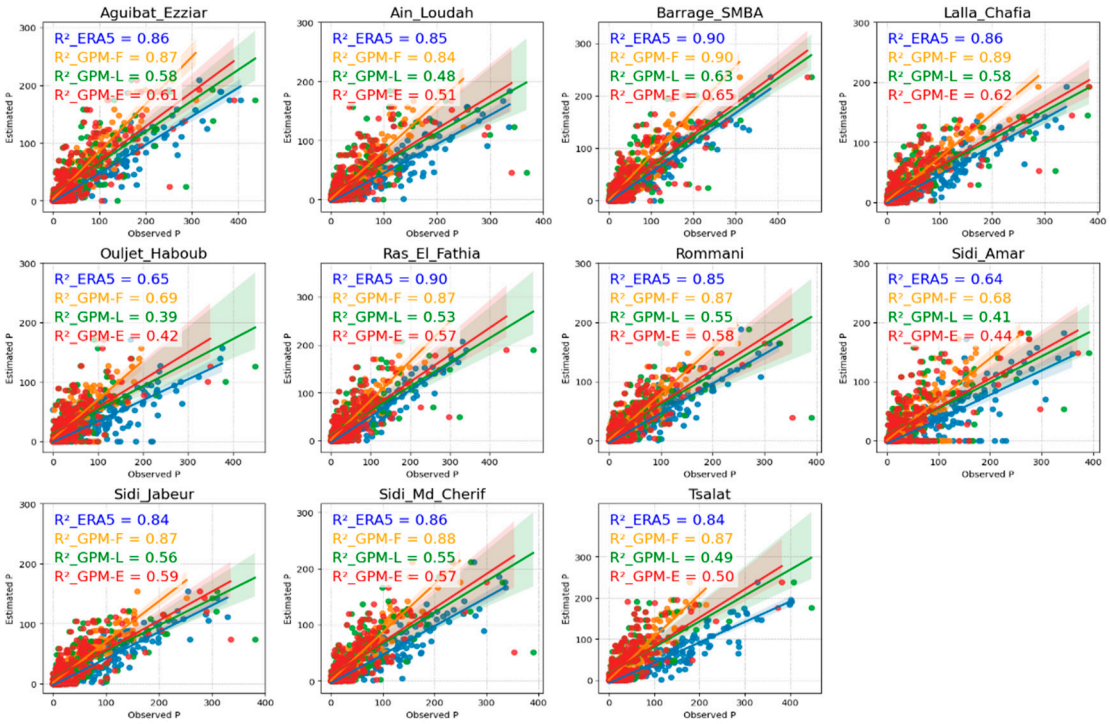


Figure 6. Linear regression between the monthly precipitation observed by the rain gauges and that estimated by the precipitation products IMERG and ERA5.

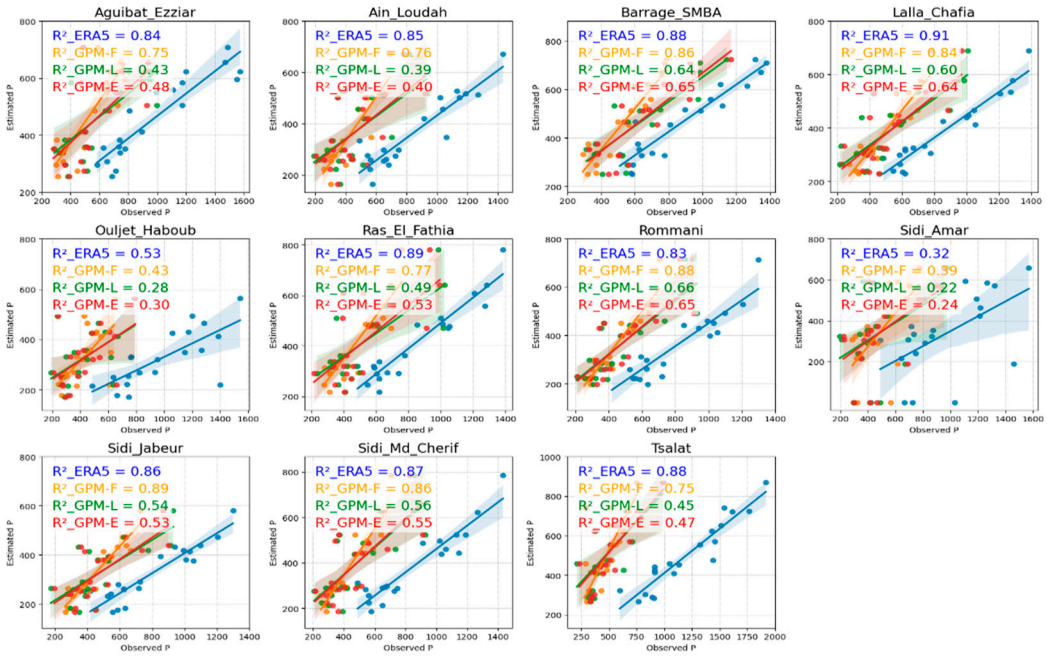


Figure 7. Linear regression between the annual precipitation observed by the rain gauges and that estimated by the precipitation products IMERG and ERA5.

3.2. Evolution of Metrics According to Altitude

The analysis of the effect of altitude on the performance of the GPPs reveals interesting findings. It should be noted that altitudes in the study area range from 11 to 1624 m, and the rain gauge altitudes vary between 67 and 687 m. Therefore, there is no clear impact of altitude on the classification metrics (Figure 8). However, when considering the regression metrics based on altitude, the ERA5 product showed a notable increase in bias at higher altitudes (Figure 9). ERA5 tends to overestimate observed precipitation in high-altitude

stations and exhibits a moderate overestimation in low-altitude stations. This suggests that the ERA5 product is more sensitive to altitude, leading to deviations in precipitation estimates at different altitudes. On the other hand, the GPM products (GPM-F, GPM-L, and GPM-E) do not show a significant effect based on altitude. This implies that the GPM satellite products are less influenced by altitude variations in the study area.

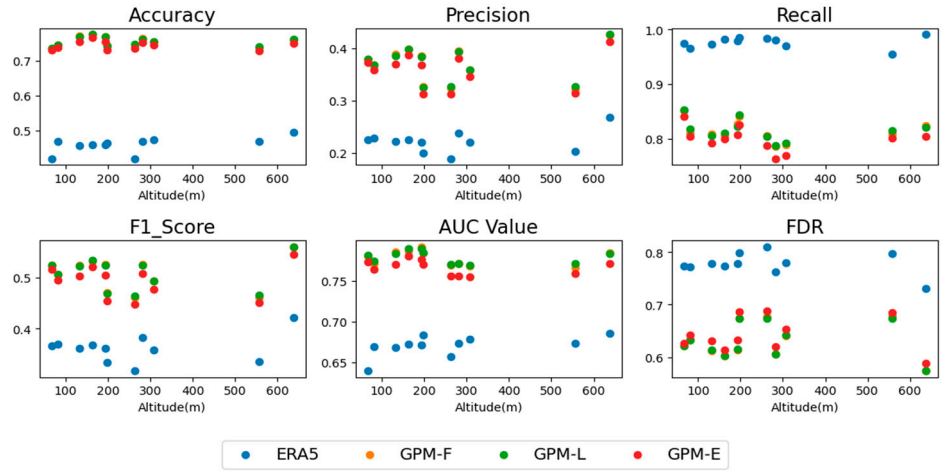


Figure 8. Evolution of classification metrics as a function of altitude.

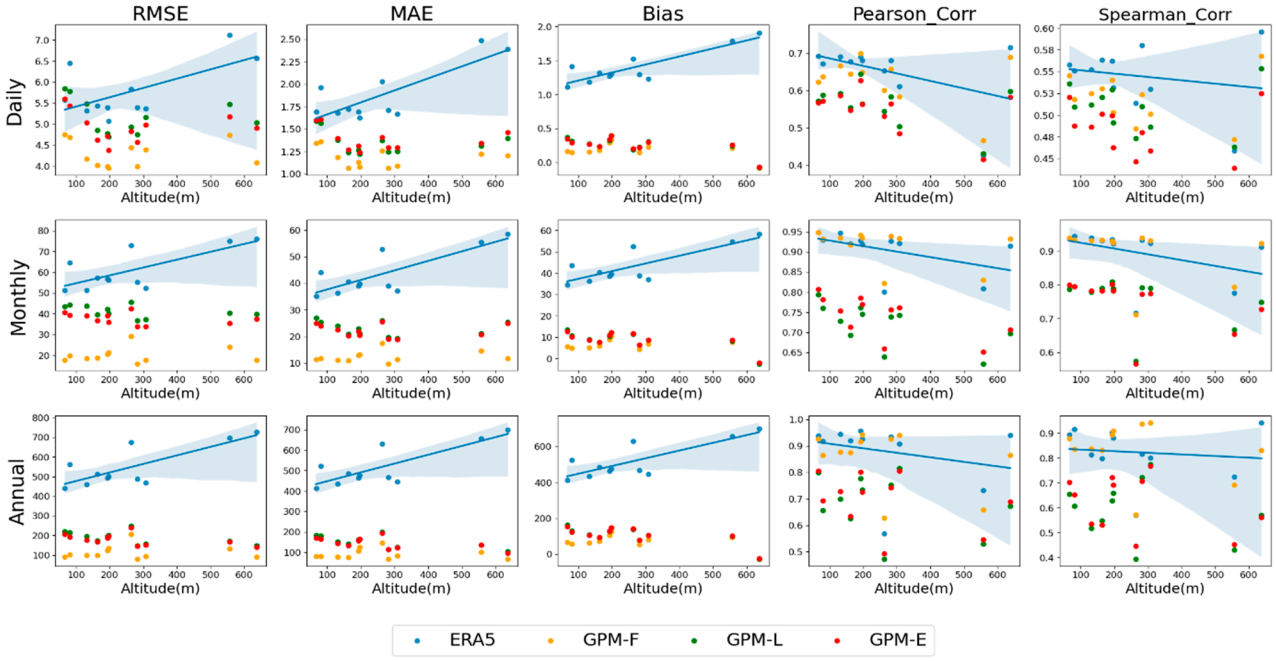


Figure 9. Evolution of the regression metrics as a function of altitude for the daily, monthly, and annual scales (Blue line: Regression line; Blue area: Confidence interval).

The overestimation of precipitation in the ERA5 and satellite products can be attributed to several factors, as supported by various studies: (a) Dynamic and thermodynamic processes: Ou et al. (2023) [79] suggest that ERA5 overestimates precipitation due to the overestimation of summer winds in the Tibetan Plateau region. Izadi et al. (2021) [80] propose a thermodynamic bias, as ERA5 overestimates atmospheric humidity in Iran, leading to an overestimation of precipitation. Qian et al. (2021) [81] found that dynamic processes such as convection and vertical air movements, along with a thermodynamic bias from overestimated humidity, contribute to the overestimation of ERA5 precipitation in the Tibetan Plateau region. (b) Spatial resolution: Kendon et al. (2019) [82] highlight that

the spatial resolution of climate models, including ERA5, can contribute to the overestimation of precipitation. Qin et al. (2020) [83] found an overestimation of precipitation in ERA5 over mainland China due to its low spatial resolution. (c) Representation of cloud and precipitation processes: Qin et al. (2020) [83] demonstrate that ERA5 overestimates precipitation over mainland China due to an inadequate representation of cloud and precipitation processes, particularly an overestimation of high-intensity events. (d) Atmospheric aerosols: Rangwala et al. (2013) [84] show that atmospheric aerosols can modify precipitation distribution and lead to an overestimation of precipitation in high-altitude areas. The results also indicate increases in both MAE and RMSE from low to high altitudes at all scales. Furthermore, the Pearson and Spearman correlation coefficients decreased as altitude increased, for both ERA5 and GPM products, with a particularly pronounced decrease at the daily scale for the ERA5 (Figure 9).

3.3. Assessment Based on Rainfall Intensity

The objective of this evaluation was to explore the performance of the IMERG and ERA5 precipitation products in detecting different precipitation intensities (Table 5). For the first class ($0 \leq T < 1$ mm), the GPM-F, GPM-L, and GPM-E models show similar performances with high median accuracy values (0.88 to 0.89) and F1_scores of 0.94 (Figure 10). This suggests that these methods are accurate in predicting low-intensity precipitation and have a good balance between recall and precision. However, the ERA5 product performs differently, exhibiting higher precision (0.98) but slightly lower recall (0.82), resulting in a lower F1_score (0.89). This indicates that the ERA5 product is more likely to correctly predict positive values but may miss some positive instances. The ERA5 product also has a lower false discovery rate (FDR) (median value of 0.02) compared to the GPM products (median value of 0.06), indicating a lower rate of false positives. However, for the remaining four classes representing higher precipitation intensities, all tested products show relatively low performances, with median accuracy values ranging from 0.1 to 0.24. The recall values are also low, indicating that the products struggle to detect positive data accurately. There is a trend of improvement in recall as the precipitation intensity increases, but the values remain relatively low for all classes. The precision values are also low, suggesting a high number of false positives. This is further confirmed by the high FDR values, indicating a high rate of false positives for all products. The F1_score, which combines precision and recall, is also low, ranging between 0.18 and 0.39 for the four classes (Figure 10).

Overall, the results show that the GPP's performance is stronger for low-intensity precipitation. The GPM products perform relatively better than the ERA5 reanalysis product in terms of overall accuracy and balance between recall and precision for low-intensity precipitation. However, all products face challenges in accurately predicting and detecting higher-intensity precipitation events.

The regression metrics provide insights into the performance of the GPPs in terms of bias, RMSE, MAE, and correlation coefficients for different precipitation intensity classes. Regarding bias, all products overestimate precipitation for the first class ($0 \leq T < 1$). The GPM models have a lower bias compared to ERA5 (median of 0.03 and 0.07, respectively). For the second class ($1 \leq T < 5$), ERA5 overestimates precipitation (median bias of 0.32), while the GPM models underestimate it, with median biases ranging from -0.19 to -0.25. For classes 3 and 4 ($5 \leq T < 20$ and $20 \leq T < 40$), ERA5, GPM-F, and GPM-L show an overestimation of precipitation, while GPM-E underestimates it. For the last class ($T \geq 40$), all products overestimate precipitation, with GPM-F having the lowest bias. In terms of RMSE and MAE errors, GPM-E and GPM-L models have the lowest values for classes 1 and 2, followed closely by GPM-F and ERA5. GPM-L has the lowest errors for class 3, GPM-E for class 4, and GPM-F for the last class. The results indicate that RMSE and MAE errors, as well as bias interval variations, increase from the first class to the last class for all products, indicating a decrease in accuracy as precipitation intensity increases (Figure 11).

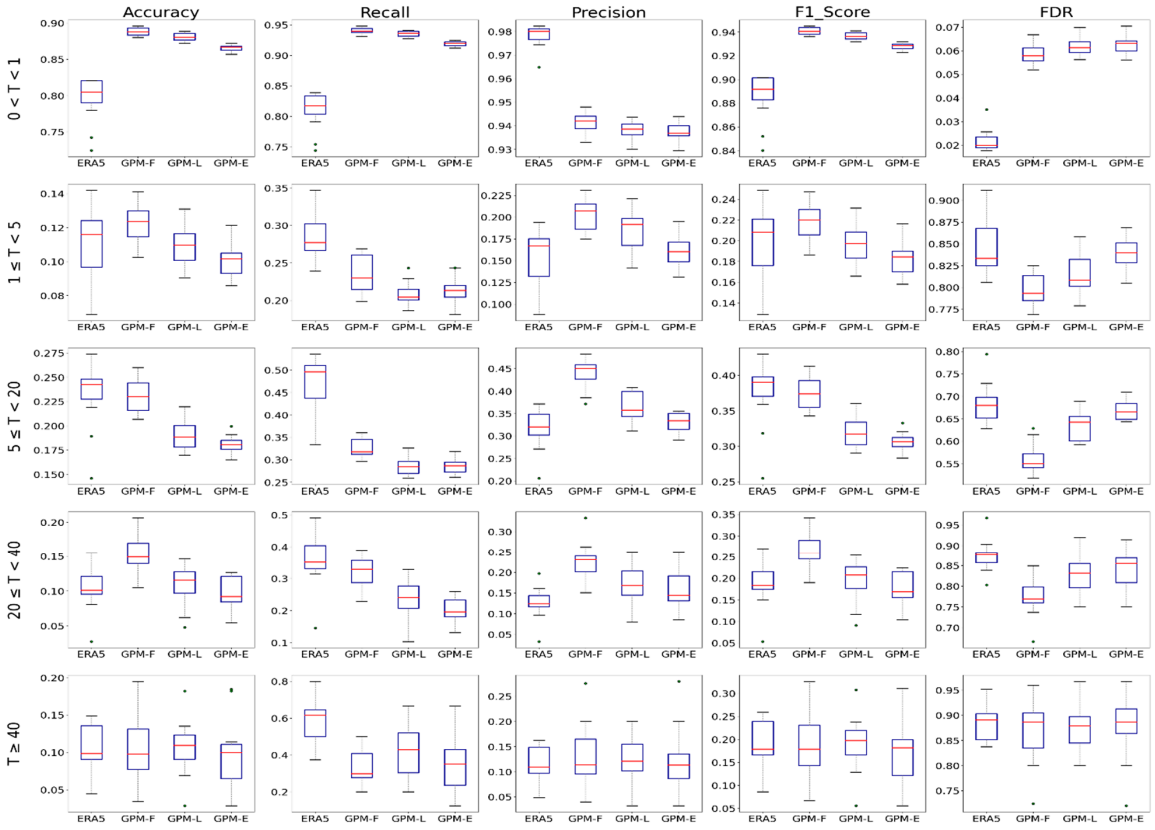


Figure 10. Boxplots of classification metrics for the different intensity classes (Red line: Median; Dots: Outliers).

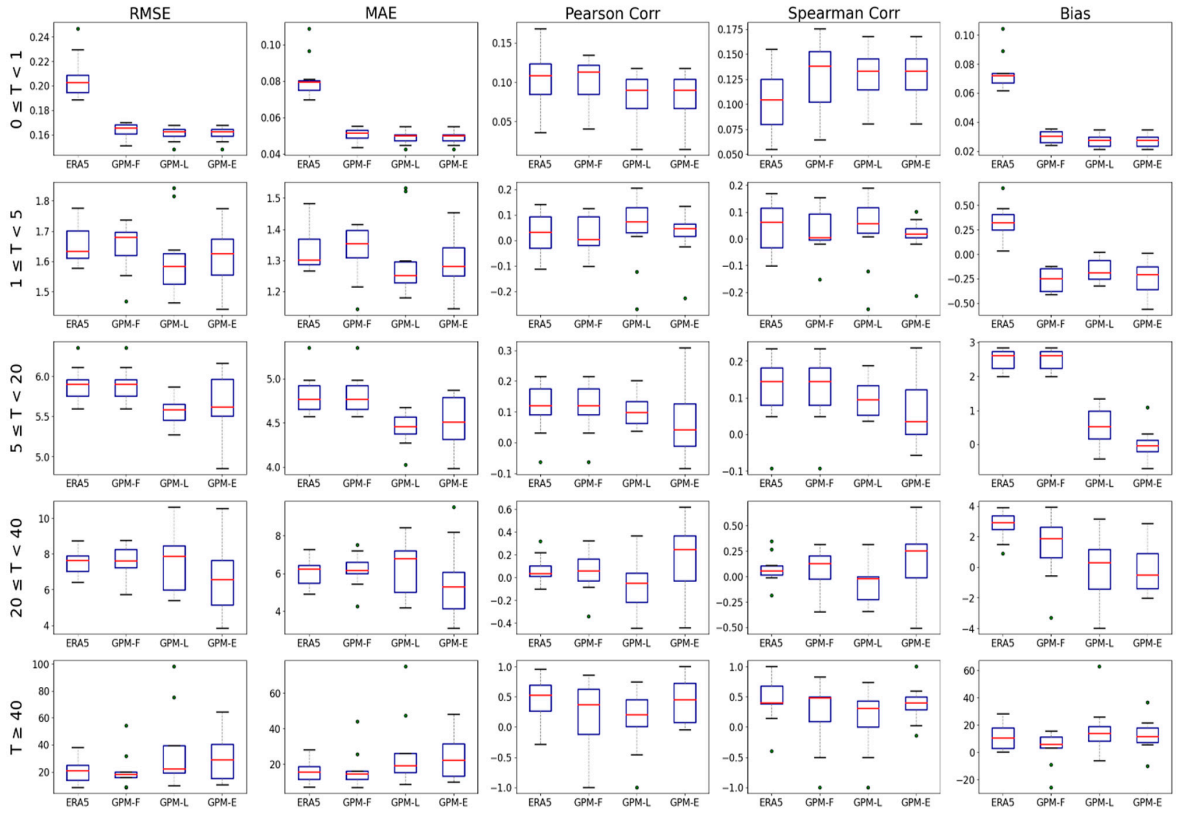


Figure 11. Boxplots of regression metrics for the different intensity classes (Red line: Median; Dots: Outliers).

When examining correlation coefficients, very low values ranging from 0.00 to 0.25 were observed for all products, except for the last class where some improvement in consistency was observed (Figure 11). This indicates that the products have limited ability to capture the spatial patterns of precipitation for the earlier intensity classes, but show some improvement for the last class.

These findings are consistent with a previous study conducted in Saudi Arabia [85], which also assessed the performance of different precipitation products from the GPM-IMERG system in an arid region with diverse topography and varying rainfall intensities. The study reported similarly low correlation coefficients for all products, with improvements observed only for the highest-intensity class.

The analysis of rainfall intensity categories and their corresponding frequencies revealed that the GPM models were able to reproduce frequencies comparable to the ground observations, while ERA5 struggled to accurately capture the frequencies for each class across all rain gauges (Figure 12). This suggests that the GPM models better reflected the distribution of rainfall intensities in the study area compared to ERA5. Figure 12 also indicated that the class ($0 \leq T < 1$) was predominant in the study area, representing over 80% of the precipitation. This highlights the importance of accurately capturing low-intensity precipitation, as it constitutes a significant portion of the total rainfall in semi-arid and arid regions.

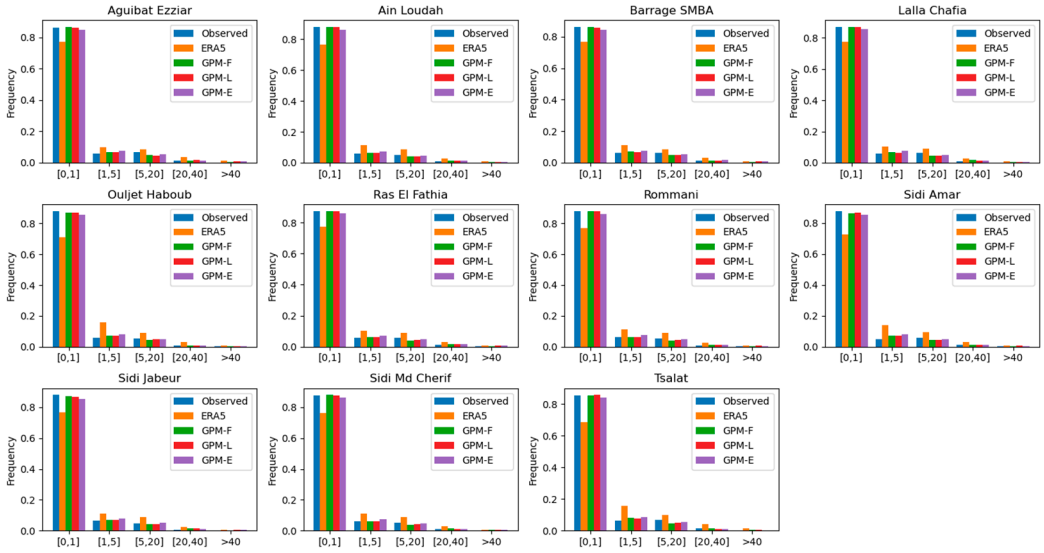


Figure 12. Frequencies of the different categories of intensities for each precipitation product and each rainfall station.

It is worth noting that as the intensity of precipitation increased, the errors in the precipitation products also increased. This suggests that the models had more difficulty accurately representing higher-intensity rainfall events. This finding aligns with the general understanding that estimating high-intensity precipitation is more challenging due to the complex dynamics and localized nature of such events.

3.4. Assessment Based on Rainy Months and Summer Months

The current study conducted a seasonal evaluation based on the rainy and summer months. In the study region, the wet season coincides with the cold season, typically spanning from November to March. The summer season is the dry season, and summer rains, usually in the form of thunderstorms, typically account for less than 10% of the total annual precipitation. Figure 13 shows the seasonal cycles of the GPM products compared to those of rain gauges. These cycle series were obtained by taking the arithmetic average of the multi-year mean precipitation for each month across all stations. Overall, all products were able to reproduce the seasonal cycle, with better performance for the GPM-F product and difficulties for the GPM-L and GPM-E in reproducing this cycle. However, they were less accurate in describing

the magnitude of rainfall. The ERA5 product exhibited high magnitude, which explains why it consistently showed the largest biases, RMSE, and MAE errors.

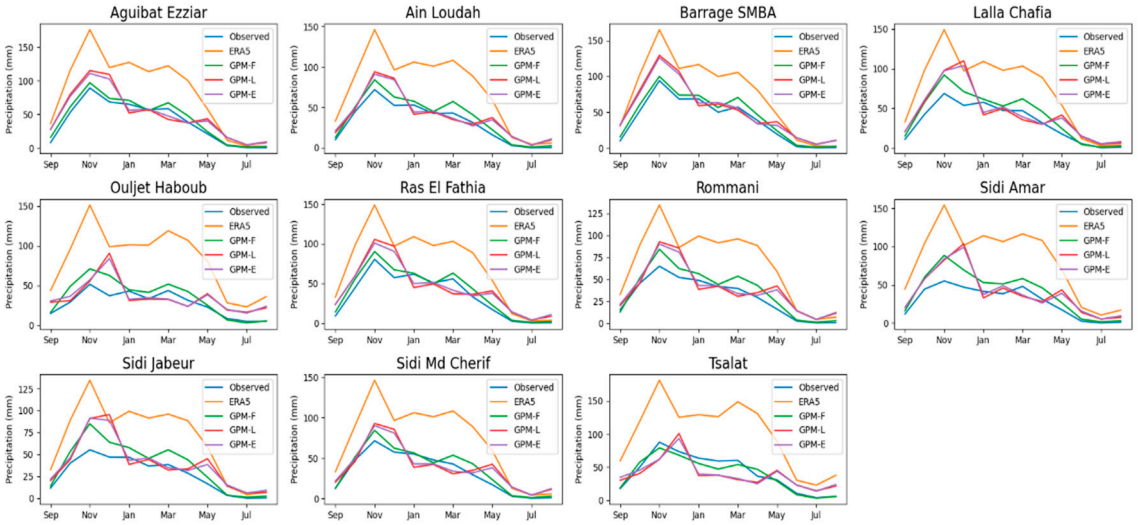


Figure 13. Ability of precipitation products to describe seasonal cycles for each rain gauge.

According to the results presented in Figure 14, all products tend to overestimate monthly precipitation from October to December, except for GPM-L, which slightly underestimates them in October. This overestimation persists for GPM-F and ERA5 during the subsequent rainy months (from January to April), while the other two GPM models (GPM-L and GPM-E) show a slight underestimation for these months. In general, ERA5 exhibits large overestimations, while the GPM products have relatively low biases. In terms of consistency, the results also revealed that the ERA5 had the highest correlation with the observed data for all rainy months, followed by GPM-F, GPM-L, and GPM-E, respectively. Similarly, in terms of the Spearman correlation coefficient, ERA5 demonstrated a better ability to capture the spatial patterns of monthly precipitation for rainy months, followed by GPM-F, GPM-L, and GPM-E. But despite its good correlations, ERA5 has the highest RMSE and MAE among the four precipitation products. The results of the RMSE and MAE also indicate that the GPM-F produced the lowest errors for the months of October to December, while from January to April, the lowest errors were obtained with the GPM-L model (Figure 14).

The results displayed in Figure 15 indicate that the GPM-F product had the lowest bias in the summer months, followed by GPM-L and GPM-E. However, ERA5 presents low performance, except for the month of August where it outperforms the GPM-L and GPM-E products. The RMSE and MAE were also lower for GPM-F, followed by GPM-E, GPM-L, and ERA5, respectively. The values of Pearson and Spearman correlation were relatively low. Overall, the precipitation products exhibit low biases as well as low RMSE and MAE for the summer months compared to the rainy months. However, they show higher correlations for the rainy months than for the summer months.

Figures 16 and 17 present the variation of observed average daily precipitation as well as precipitation products during the rainy and summer months from 2000 to 2020. In general, none of the evaluated models are able to satisfactorily reproduce the average daily precipitation. However, when comparing the four products, it is observed that the GPM models (especially GPM-F) exhibit better capability than ERA5. By examining the frequencies of different rainfall intensities for each month and rain gauge, similar results to those presented in Figure 12 were obtained to assess the products' ability to reflect these frequencies. The comparison revealed that the GPM products were able to successfully reproduce the frequencies of all five classes, while ERA5 tended to underestimate the lowest rainfall class ($0 \leq T < 1$) and overestimate the other classes.



Figure 14. Boxplots of rainy month regression metrics (Red line: Median; Dots: Outliers).

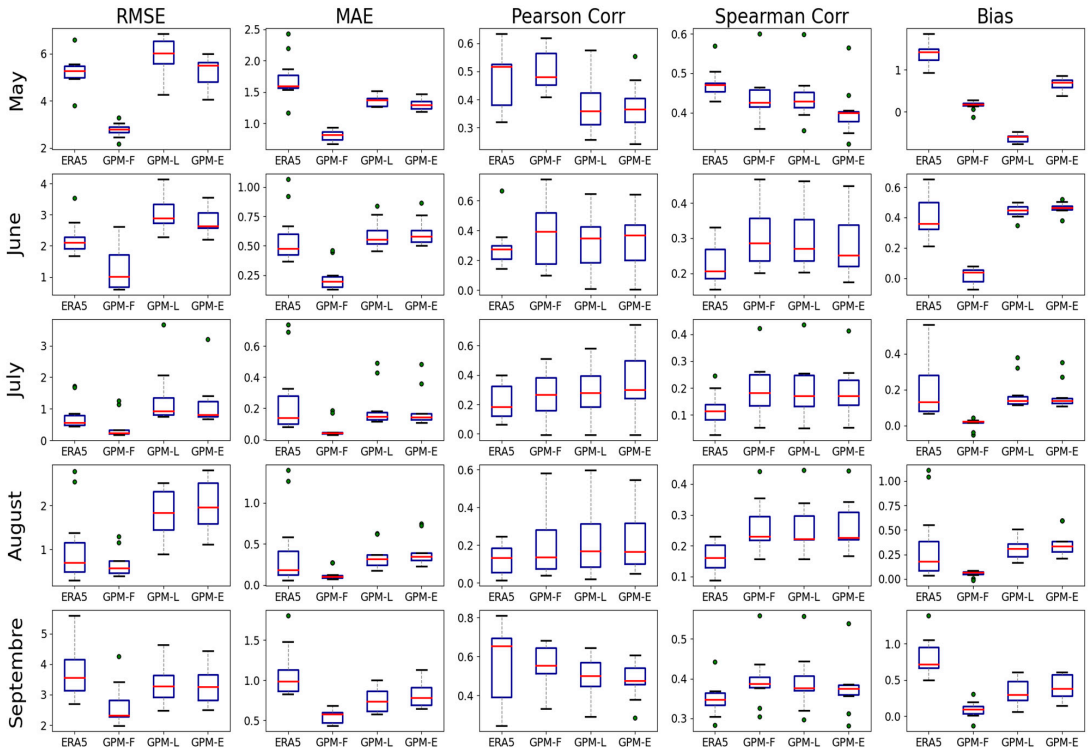


Figure 15. Boxplots of summer months regression metrics (Red line: Median; Dots: Outliers).



Figure 16. Ability of precipitation products to describe the average daily precipitation of the rainy months for each station.

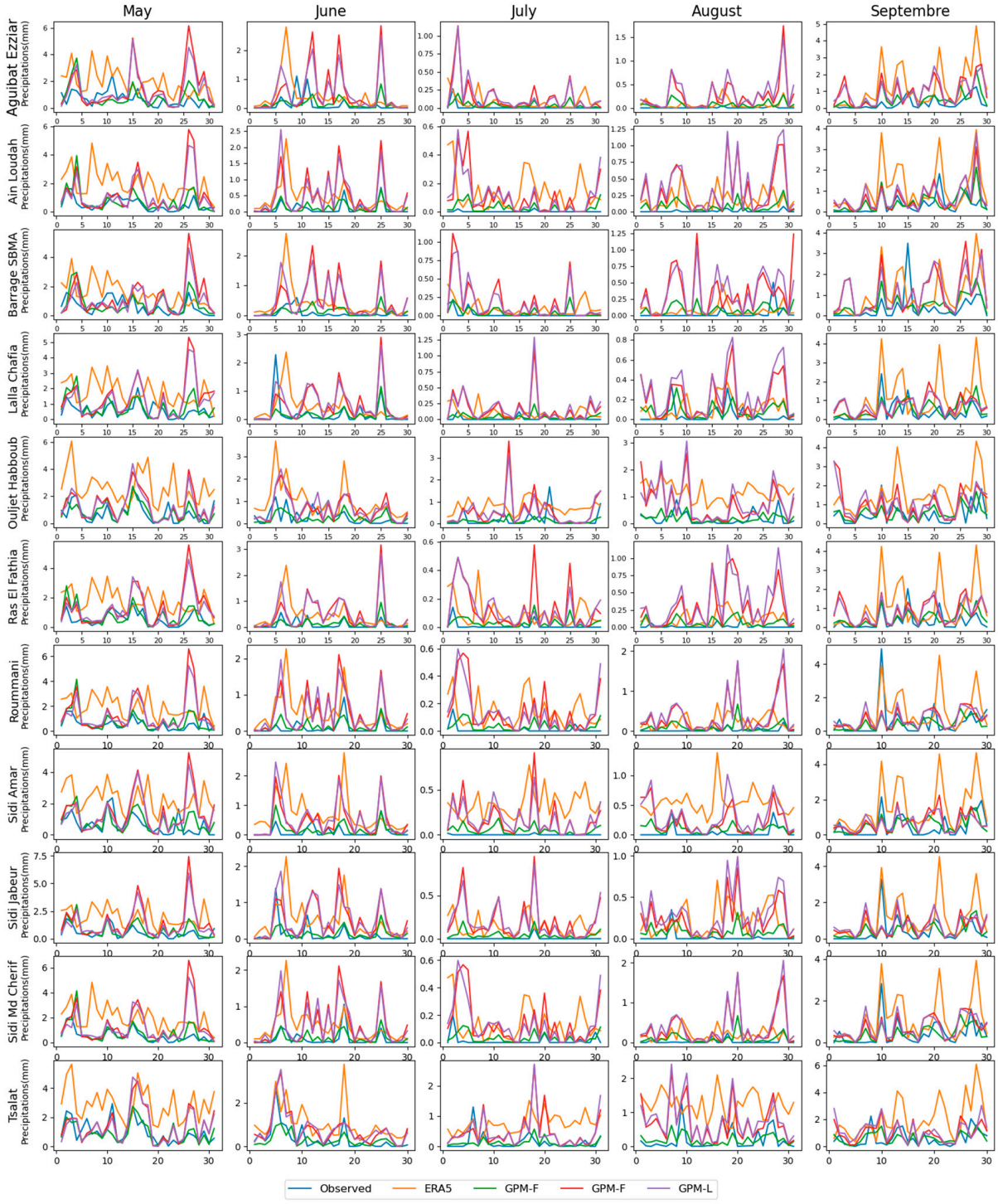
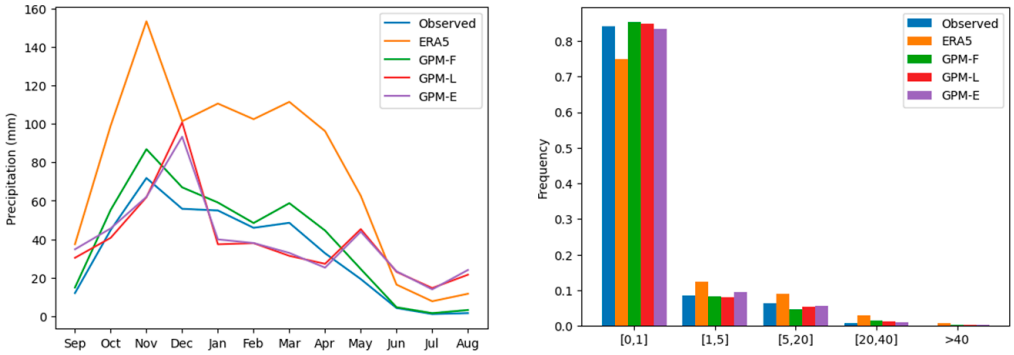


Figure 17. Ability of precipitation products to describe the average daily precipitation of the summer months for each station.

3.5. Evaluation Based on Pixel Average

As part of this study, an evaluation was conducted using the average of rain gauge stations and the average of their pixels for all precipitation products. The GPM-F model best describes the seasonal cycle among the four products (Figure 18a). The GPM-F model showed superior performance compared to other products most likely due to advancements in calibration techniques and validation methodologies that help refine the accuracy of satellite precipitation products. GPM final benefits from these advancements, which may involve more accurate radiometric calibration, better accounting for sensor biases, and

enhanced validation against ground-based observations [7,8,39]. In terms of magnitude, ERA5 overestimates precipitation excessively compared to other products, resulting in high amplitude (Figure 18a). Furthermore, it is visible that the GPM-L and GPM-E models underestimate precipitation for the months of January to April, which is consistent with the regression metric results for rainy months (Figure 14).



(**a**) (**b**)

Figure 18. (**a**) Ability of each product to describe precipitation cycles for the pixel average. (**b**) Frequencies of the different intensity categories of each precipitation product for the pixel average.

Since the beginning of the study, the GPM-F model has shown the lowest bias for all temporal scales, followed by GPM-L, GPM-E, and ERA5. The RMSE and MAE are also lowest with the GPM-F model, followed by GPM-E, GPM-L, and ERA5. In terms of consistency, ERA5 remains the most correlated product with ground data, followed by GPM-F, GPM-L, and GPM-E (Figure 19).

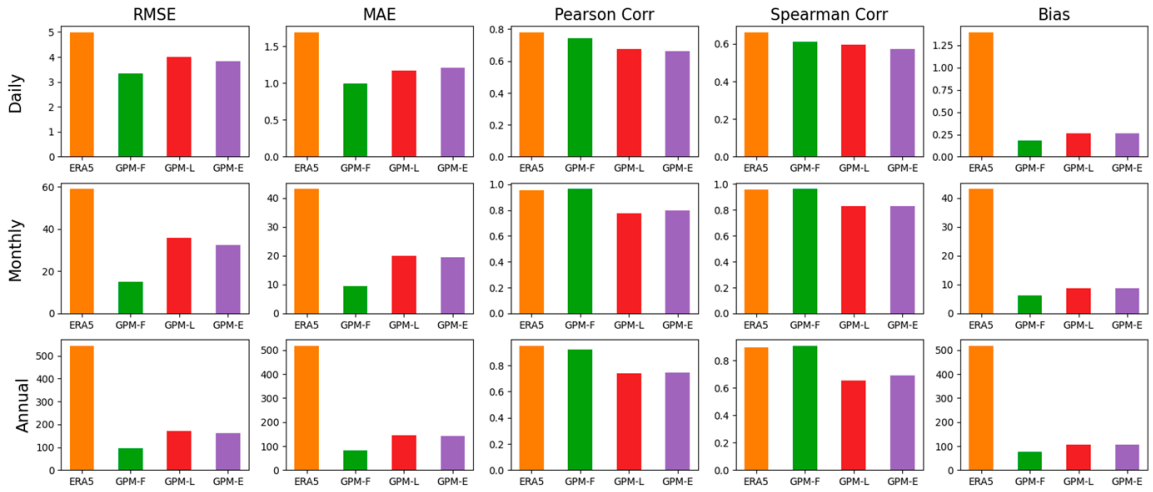


Figure 19. Pixel mean regression metric bar plots for all precipitation products.

The ability of precipitation products to describe daily average precipitation remains generally unsatisfactory, but the GPM-F performs better than the other products (Figure 20). In terms of frequency, the GPM products are able to reflect the frequencies of each class. However, ERA5 tends to underestimate the frequencies of the class ($0 \leq T < 1$), which represents the majority of precipitation in the study area, and overestimate the frequencies of other classes (Figures 21 and 18b).

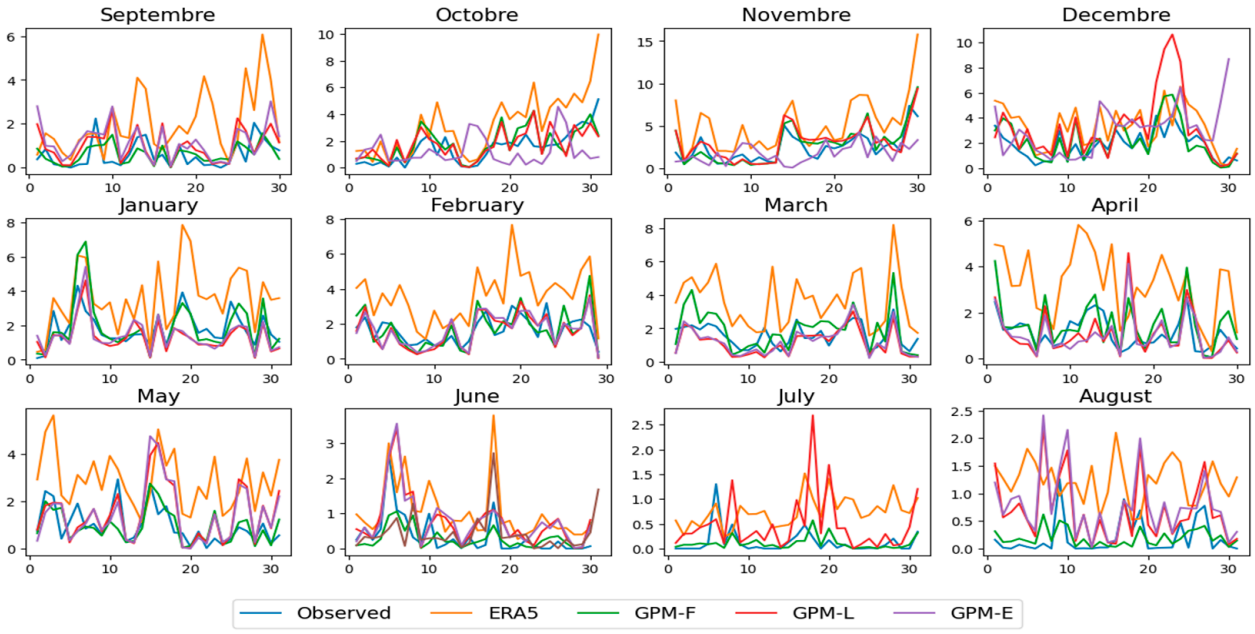


Figure 20. Ability of each product to describe average daily precipitation for the pixel average.

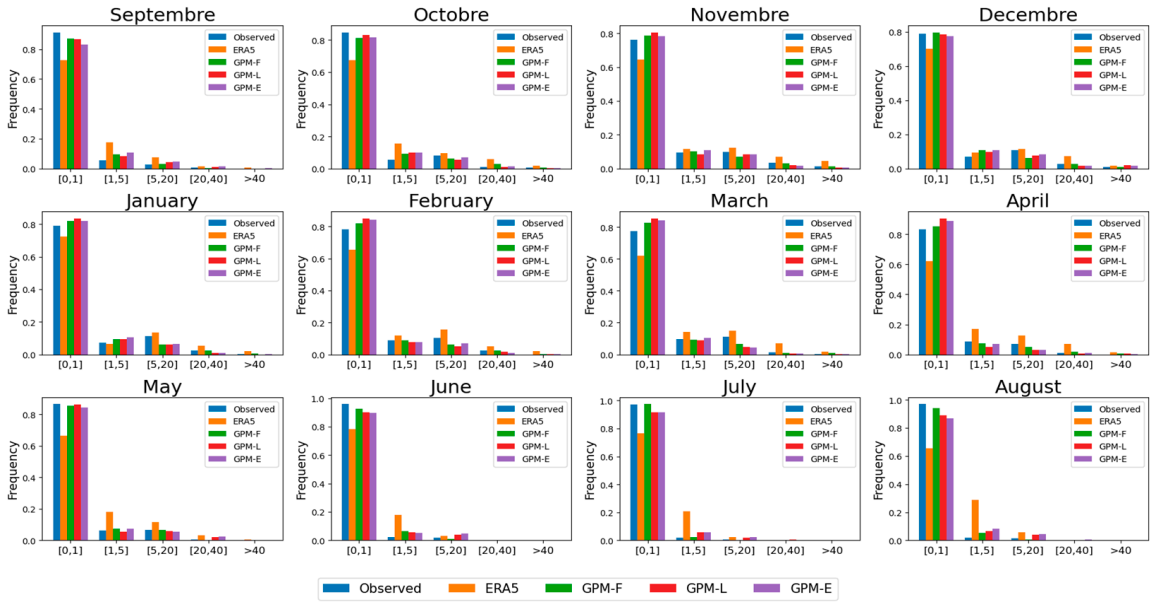


Figure 21. Frequencies of the different intensity categories for each month and each precipitation product for the pixel average.

3.6. Bias Correction Results

According to the results obtained from regression and classification metrics, the GPM-F product outperforms all the other GPPs at various scales. Previous studies have also shown that the GPM-F product outperforms other products. For example, in Saudi Arabia, [86] compared the three versions of the IMERG product and concluded that GPM-F was the most accurate and could be used in conjunction with ground measurements. In Iran, ref. [87] evaluated four products and found that GPM IMERG and TRMM were more accurate than Chirps in estimating precipitation amounts. Similar results were found in China, Malaysia, and Pakistan by [8,88,89] using the GPM-F product. Furthermore, Rachdane et al. [19] evaluated six satellite products in sub-Saharan Morocco and demonstrated that GPM-F had the highest correlations and the lowest MAE and RMSE for all time scales.

In order to improve the accuracy of the GPM-F model, we have applied a bias correction. Figures 22 and 23 illustrate the regression and classification metrics of the GPM-F model before and after the bias correction. These results are derived from a test conducted on the artificial neural network (ANN) model constructed and trained for this study. Following the correction, the results show a significant reduction in biases and in RMSE and MAE at all time scales. Additionally, there is a slight improvement in the coefficients of determination and correlation at the daily and annual scales, but insignificant improvement at the monthly scale.

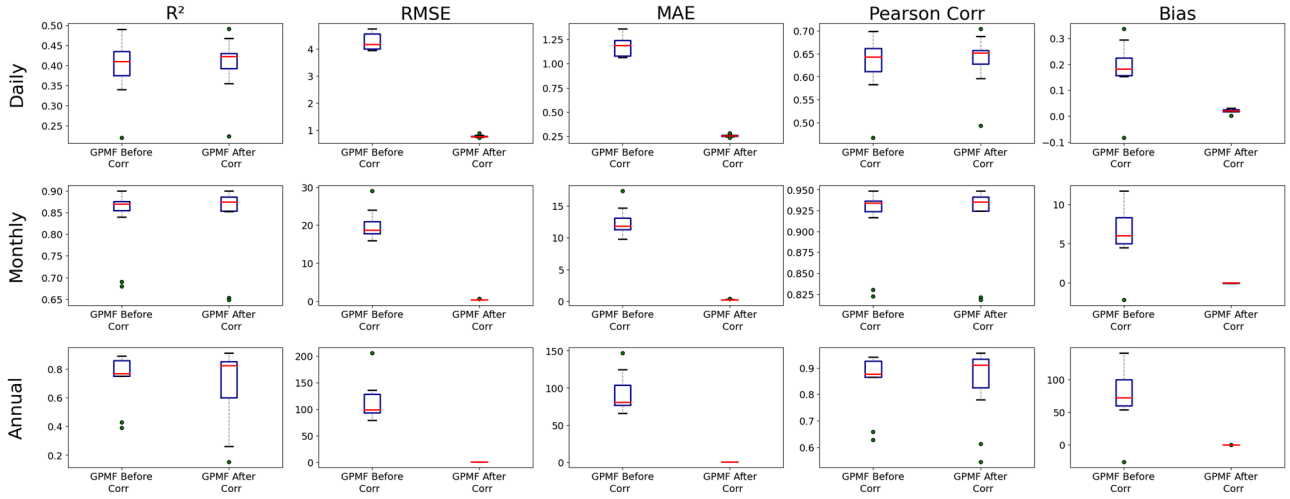


Figure 22. Boxplots of regression metrics before and after bias correction (Red line: Median; Dots: Outliers).

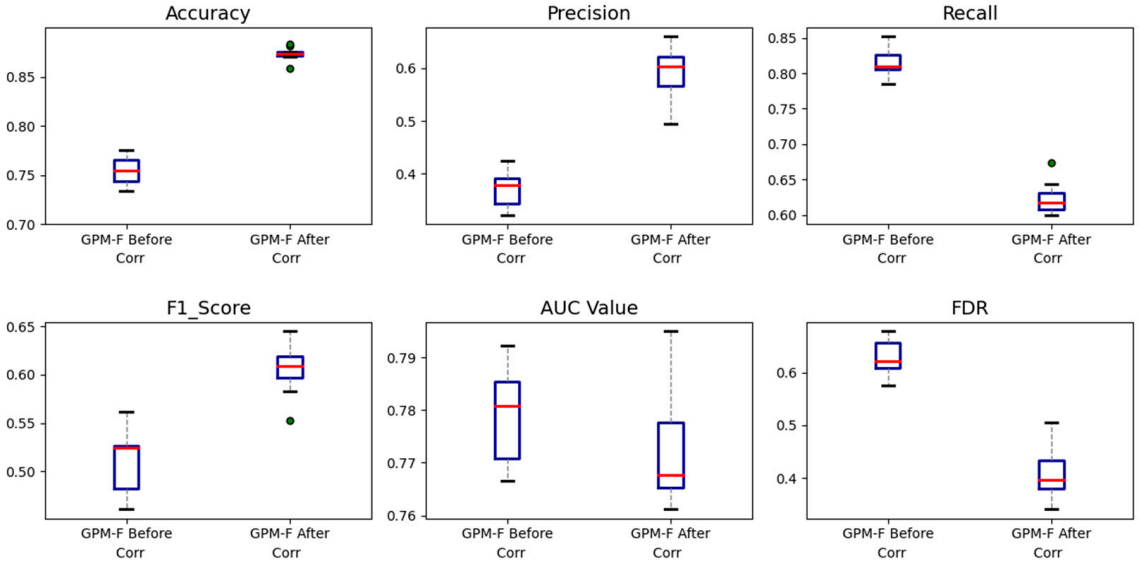


Figure 23. Boxplots of classification metrics before and after bias correction (Red line: Median; Dots: Outliers).

At the daily scale, the correction resulted in a significant decrease in the median bias, from 0.18 to 0.02, as well as the median RMSE and MAE, from 4.17 and 1.19 to 0.76 and 0.25, respectively. The correlation increased from 0.64 to 0.65, and R^2 increased from 0.41 to 0.42. At the monthly scale, the correction also led to a significant decrease in the median bias, from 6.04 to 0.012, as well as the median RMSE and MAE, from 18.71 and 11.83 to 0.35 and 0.93, respectively. However, no significant improvement in correlation was observed at this scale. At the annual scale, the correction resulted in a considerable reduction in the median bias, from 72.44 to 0.02, as well as the median RMSE and MAE, from 98.86 and 80.64 to 0.42

and 0.36, respectively. The correlation increased from 0.88 to 0.91, and R^2 increased from 0.77 to 0.83 (Figure 22).

After the bias correction, the classification metrics showed a significant improvement in the overall performance of the model, with an increase in accuracy from 0.75 to 0.87, indicating a better ability to correctly predict the correct class for each observation. Precision also increased from 0.38 to 0.60, suggesting that the model is capable of predicting the positive class with greater accuracy and identifying true positives more precisely. However, recall decreased from 0.81 to 0.61, indicating that the model has a lower ability to identify all true positives, and the number of false negatives increased. The F1_score increased from 0.52 to 0.60, indicating an overall improved performance in terms of precision and recall. However, the AUC-ROC decreased from 0.78 to 0.76, suggesting a slight reduction in the model's ability to discriminate between positive and negative classes. Finally, the FDR decreased from 0.62 to 0.4, indicating a decrease in the percentage of incorrect positive predictions, demonstrating the model's improved ability to minimize false positives (Figure 23).

4. Conclusions and Perspectives

The Bouregreg watershed serves as a representative example of an arid and semi-arid region characterized by high spatial and temporal variability in precipitation. Being situated in the Mediterranean region, this area is especially susceptible to climate change impacts. Accurate precipitation data are crucial for the region's development, considering its significant influence on various aspects. However, due to the limited number of measurement stations, reanalysis and global satellite data with continuous spatial coverage play a crucial role. Among the available products, IMERG and ERA5 represent modern gridded precipitation datasets that offer high-quality records, fine resolutions, and extensive spatiotemporal coverage, making them valuable for water resource management. Therefore, it is essential to thoroughly evaluate their performance in the Bouregreg watershed. In this regard, this study conducted a comprehensive assessment of three IMERG products (GPM-F, GPM-L, and GPM-E) and the ERA5 reanalysis product, utilizing ground-based daily observations from September 2000 to August 2020 in the Bouregreg watershed.

The key findings of the study indicate that the GPM-F product exhibited the best performance across various temporal and spatial scales, followed by the GPM-L and GPM-E products. On the other hand, the ERA5 reanalysis product displayed the weakest performance at pixel evaluation. In terms of spatial analysis, the GPM models did not show significant variation with altitude, likely due to the relatively low altitude range of the rain gauges in the study area, which does not exceed 620 m. Conversely, the ERA5 product demonstrated lower performance at higher altitudes, characterized by increasing bias and decreasing correlation with increasing altitude. Additionally, the GPM products demonstrated good detection capability, precision, and accuracy at both temporal and spatial scales. All products, particularly GPM-F, showed favorable performance for precipitation intensities below 1 mm per day. In contrast, ERA5 exhibited better detection capability for higher intensities ($\geq$40 mm/day). Overall, all four datasets successfully reproduced seasonal cycles and spatial distributions, although GPM-F exhibited superior performance in capturing seasonal cycles, while ERA5 displayed high consistency in spatial distribution. Moreover, all products exhibited high biases during the rainy months and low biases during the summer months.

In conclusion, the results suggest that precipitation products, especially the GPM-F product, are well-suited for arid and semi-arid environments and can be utilized for effective water resource quantification and management. Enhancing event detection capability and accuracy is crucial for improving gridded precipitation products. After applying the bias correction using the ANN model, the GPM-F product showed significant improvements in several daily precipitation indices, including accuracy, precision, F1_Score, and FDR. Additionally, the precision of the GPM-F in terms of bias, RMSE, and MAE also increased. However, although the accuracy of the GPM-F product has been improved,

there is still potential for further enhancement, particularly in terms of recall, AUC value, and coefficient of determination. This would result in more accurate and reliable rainfall estimates for the Bouregreg watershed and similar regions.

Finally, there is no doubt that it is interesting to assess the models' performance in other regions. On this subject, we point out the importance of conducting additional research and validation in order to generalize our findings beyond the studied basin and expand the use of gridded data to other hydrological applications such as surface water management, hydrological modeling, or the monitoring of drought and floods.

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